بِسْمِ ٱللَّهِ ٱلرَّحْمَـٰنِ ٱلرَّحِيمِ

Bio_203

CURRENT MID_TERM PAPERS

Fall 2022_23

Full Solved

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ZOOLOGY
LEGENDS

"Don't Stop When you Tired, Stop When you Done"

0304-4756496

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Mcqs:

(Mostly from Files)

Short Questions: (Most Ans are Taken from PPTs)

(Repeated)

1. dNTPs

Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification.

dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 μM.

2. Nested and Multiplex PCR

NESTED PCR	MULTIPLEX PCR
Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR. Very low probability of nonspecific amplification.	Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences

3. In situ PCR

 In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide.

 In situ PCR amplification can be performed on fixed tissue or cells.

 Applies the methodology of hybridization of the nucleic acids.

 Allows identification of cellular markers

 Limited to detection of non-genomic material such as RNA.

4. Type 1 of Restriction and Modification (R-M)

Type I

♣ Type I systems were the first to be characterized from E. coli K 12

♣ The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit

♣ Type I systems are of little value for gene manipulation

5. Negative control

 Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

 The PCR product should be of the expected size.

6. Nomenclature of The Smith and Nathan's (1973)

♣ A suitable system was proposed by Smith and Nathans (1973)

♣ The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet

♣ E. coli = Eco

♣ H. influenzae = Hin

♣ Strain identification is written as Eco K

♣ In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of H. influenzae

♣ HindI , HindII

7. dNTPs or Assembly PCR:

Deoxynucleoside triphosphates (dNTPs)	**ASSEMBLY PCR**
Deoxynucleoside triphosphates (dNTPs) are the building blocks from which the DNA polymerase synthesizes a new DNA strand during successive cycles of PCR amplification. dNTPs consist of four basic nucleotides - dATP, dCTP, dGTP, and dTTP.The concentration of each dNTP (dATP, dCTP, dGTP, dTTP) in the PCR is usually 200 μM.	□ Assembly PCR used to assemble two or more pieces of DNA into one piece. Assembly PCR **PRINCIPLE** □ Assembly PCR is the synthesis of long DNA structures by performing PCR on a pool of long oligonucleotides with short overlapping segments, to assemble two or more pieces of DNA into one piece.

8. Application of PCR in microbiology.

□ Identification of bacteria:

□ Slow growing bacteria

□ Bacterial antibiotic resistance genes.

□ VIRUS: almost all viruses identification and viral load.

□ FUNGUS

□ Aspergillus spp

□ Candida spp

□ Plasmodium spp

□ Trypanosoma spp

9. Steps of PCR

THREE STEPS

□ Denaturation

□ Annealing

□ Extension S

10. Nested primers.

Nested primers (as they lie / are nested within the first fragment).

11. Asymmetric PCR

 Asymmetric PCR is used to amplify one strand of the original DNA more than the other.

 It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.

 PCR is carried out as usual, but with a great excess of one primers for the chosen strand.

 Due to the slow amplification later in the reaction after the limiting primer has been used up, extra cycles of PCR are required.

12. Advantages of PCR

 Specific

 Simple, rapid, relatively inexpensive

 Amplifies from low quantities

 Works on damaged DNA

 Sensitive

 Flexible

13. Types of Restriction and modification (R-M) system

Types of restriction and modification (R-M) system

♣ At least **four** R-M systems are known

I. Type I
II. Type II
III. Type III
IV. Type IIs

14. 3 Steps of designing Primers (Ans from Google)

How to Design Primers for PCR and quantitative real time PCR (qPCR)

- Keep the melting temperatures (T_m) of each primer pair within 2°C of one another
- Use an annealing temperature (T_a) 3-5°C below the melting temperature
- Keep primers between 18-22 base pairs long
- Design primers with a GC content of 35-65%.

15. Ligation Mediated PCR

 Uses small DNA oligonucleotide 'linkers' (or adaptors) that are first ligated to fragments of the target DNA.

 PCR primers that anneal to the linker sequences are then used to amplify the target fragments.

 DNA sequencing

 Genome walking

 DNA foot printing

16. Methodology of Assembly PCR

 It involves an initial PCR with primers that have an overlap and a second PCR using the products as the template that generates the final full-length product.

 Assembly PCR used to assemble two or more pieces of DNA into one piece.

17. What reagents are used in standard PCR? (3)

Generally, a complete polymerase chain reaction (PCR) requires five basic reagents, including

- DNA template,
- Forward and reverse primers,
- DNA polymerase,

- ♣ Deoxy nucleotide triphosphates (dNTPs)
- ♣ Reaction buffer.

18. Uses of Asymmetric PCR

 It finds use in some types of sequencing and hybridization probing where having only one of the two complementary strands is required.

Long Questions: (Repeated)

1. Microbiology application:

 Identification of bacteria:

 Slow growing bacteria

 Bacterial antibiotic resistance genes.

 VIRUS: almost all viruses identification and viral load.

 FUNGUS

 Aspergillus spp

 Candida spp

 Plasmodium spp

 Trypanosoma spp

2. Forensic Application of PCR

 Forensic applications: DNA finger printing
 Detection of Mutations: Inherited disorders & carriers Track DNA abnormalities
 Prenatal diagnosis of genetic disorders.
 Detection of pathogens Pre-natal diagnosis DNA fingerprinting Gene therapy Mutation screening
 Drug discovery
 Classification of organisms Genotyping

3. Nested PCR differ Multiplex PCR

NESTED PCR	MULTIPLEX PCR
Two pairs (instead of one pair) of PCR primers are used to amplify a fragment. First pair amplify a fragment similar to a standard PCR. Very low probability of nonspecific amplification.	Multiplex PCR is a variant of PCR which enables simultaneous amplification of many targets of interest in one reaction by using more than one pair of primers. The use of multiple, unique primer sets within a single PCR reaction to produce amplicons of varying sizes specific to different DNA sequences

4. Amplified fragment length polymorphism (AFLP) PCR

METHODOLOGY

- Genomic DNA is digested with one or more restriction enzymes. tetracutter (MseI) and a hexacutter (EcoRI).
- Ligation of linkers to all restriction fragments. Pre-selective PCR is performed using primers which match the linkers and restriction site specific sequences.
- Electrophoretic separation and amplicons on a gel matrix, followed by visualization of the band pattern.

AFLP PCR

- AFLP is a highly sensitive PCR-based method for detecting polymorphisms in DNA.
- AFLP can be also used for genotyping individuals for a large number of loci.

5. Hot start PCR

- This is a technique that reduces nonspecific amplification during the initial set up stages of the PCR.

- The technique may be performed manually by heating the reaction components upto the melting temperature (e.g, 95°C).

- Before adding the polymerase, specialized enzyme systems have been developed that inhibit the polymerase's activity at ambient temperature.

- This is done either by the binding of an antibody or by the presence of covalently bound inhibitors that only dissociate after a high-temperature activation step.

- DNA Polymerase- Eubacterial type I DNA polymerase, Pfu.

• These thermophilic DNA polymerases show a very small polymerase activity at room temperature.

6. Differentiate between the each Step of PCR

I. Denaturation	II. Annealing	III. Extension
• First regular cycling event and consists of heating the reaction to 93°C - 98°C for 20- 45 seconds. • It causes melting of DNA template yielding single strands of DNA.	• The reaction temperature is lowered to 40-65°C for 20-40 seconds. • Annealing of the primers to the single stranded DNA template.	• The temperature at this step depends on the type of DNA polymerase used. • Taq polymerase has its optimum activity temperature at 70- 80°C. • Commonly 72°C is used. • At this step the DNA polymerase synthesizes a new DNA strand complementary to the DNA template strand by adding dNTP's.

7. Mini Primer PCR:

 Mini Primer PCR uses a thermostable polymerase (S-Tbr) that can extend from short primers as short as 9 or 10 nucleotides.

 This method permits PCR targeting to smaller primer binding regions, and is used to amplify conserved DNA sequences, such as the 16S (or eukaryotic 18S) rRNA gene.

 PCR that can extend from short primers

8. Conventional PCR (Real Time PCR or q-PCR) & Ingredients

REAL TIME PCR

 Real Time PCR is a technique in which fluorophores bind to specific target regions of amplicons to produce fluorescence during PCR.

 Real Time PCR is used to measure the quantity of a PCR product.

 The fluorescence, measured in Real Time, is detected in a PCR cycler with an inbuilt filter flurometer.

 It is the method of choice to quantitatively measure starting amounts of DNA, cDNA or RNA

Quantitative real-time PCR is often confusingly known as RT-PCR (Real Time PCR) or RQPCR.

• QRT-PCR or RTQ-PCR are more appropriate contractions.

Ingredients

• QRT-PCR methods use fluorescent dyes, such as Sybr Green, or fluorophore-containing DNA probes, such as TaqMan, to measure the amount of amplified product in real time.

Q-PCR is commonly used to determine whether a DNA sequence is present in a sample and the number of its copies in the sample

9. What are the Limitations of PCR?

 Contamination risk

 Primer complexities

 Primer-binding site complexities

 Amplifies rare species

 Detection method

10. Methylation PCR (MSP)

 Methylation-specific PCR is used to identify patterns of DNA methylation at CpG islands in genomic DNA.

 Target DNA is first treated with sodium bisulfite, which converts unmethylated cytosine bases to uracil, which is complementary to adenosine in PCR primers.

 Two amplifications are then carried out on the bisulfite-treated DNA: One primer set anneals to DNA with cytosines (corresponding to methylated cytosine), and the other set anneals to DNA with uracil (corresponding to unmethylated cytosine).

 MSP used in quantitative PCR provides information about methylation state of a given CpG island.

11. Different between Colony PCR, In Situ PCR, Asymmetric PCR:

Colony PCR	In Situ PCR	Asymmetric PCR
□ The screening of bacterial or yeast clones for correct ligation or plasmid products.	□ In Situ PCR (ISH) is a polymerase chain reaction that actually takes place inside the cell on a slide. □ In situ PCR amplification can be performed on fixed tissue or cells .	□ Asymmetric PCR is used to amplify one strand of the original DNA more than the other. □ It is used in some types of sequencing and hybridization probing where having only one of the two complementary strands is ideal.

12. Negative control and blank reaction

NEGATIVE CONTROL

□ Controls for specificity of the amplification reaction contains all reagents and a DNA template lacking the target sequence.

BLANK REACTION

□ Controls for contamination contains all reagents except DNA template.

13. Define types of (R-M)

Type I

♣ Type I systems were the first to be characterized from E. coli K 12

♣ The active enzyme consists of two restriction subunit, two modification subunit and one recognition subunit

Type II

♣ Most of the useful R- M system is Type II

♣ Type II enzymes recognize a defined sequence and cut within it

Type III

♣ Type III enzymes have symmetrical recognition sequences but otherwise resemble type I systems and are of little value

Type IIs

♣ Type IIs systems have similar cofactors and structure to type II but restriction occurs at a distance from recognition site that limits their usefulness

14. Number and size of restriction fragments

The number and size of fragments generated by restriction enzyme depend on the frequency of occurrence of the target site in the DNA to be cut.

Four base recognition site occurs every 44 (256) bp

Six base recognition site occurs every 46 (4096) bp

Eight base recognition site occurs 48 (65,536) bp

Zoology legends:

https://chat.whatsapp.com/HMS1CGrZW4h7QWa1D53roe

Bio203:

https://chat.whatsapp.com/KfUfXKDlT8rAVzVkWZML8G

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Jazak Allaht